# DATA-Types-2

07 April 2026 12:25

Q) What are the Various Base Conversion Fun ns in Python?
ANS: the Various Base Conversion Functions in Python are

1) bin()
2) oct()
3) hex()

## float:

- used to store decimal (real) numbers
- example: 10.5, 0.9, -2.3
- supports arithmetic operations like addition, subtraction, etc.
- result of operations can be float

Note:Float data never allows use to Represent/Store Binary.Octal and Hexa Decimal Values;

- float also supports scientific notation.The Advantages of Scientific is that to Take Less Memory space for bigger floationg point values.
- example: 1.2e3 = 1200.0

How?
Formula :
EdvFloating Point Value=Manthisa*10^e
=1.2*10^3=1200.00

```
a = 12.3
b = 0.9

print(type(a))
print(type(b))

x = 10
y = 1.2
print(x + y)   # 11.2
```

a=0b101.0b111--------SyntaxError:Invalid decimal Literal
a=Oo13.Oo23----------SyntaxError:Invalid decimal Literal

## bool:

- used to store True or False values
- example: True, False->Both Are Keywords in the Programming Language.m
- case sensitive (true/false invalid)
- used in conditions and decision making

```
a = True
b = False
```

Example:

```
print(type(a))
print(type(b))

print(True + True)   # 2
print(True + False)  # 1
```

- internally True = 1 and False = 0

```
a=true
print(a)  #NameError: name 'true' is not defined. Did you
mean: 'True'?
```

```
//Practice Questions
>>> a=True
>>> b=False
>>> print(a+b)--------I----------------1
>>> print(a-b)
1
>>> print(2*True+True+False)
3
>>> print(2+True*3+False)
5
>>> print(0b1010-True+2*True)
11
>>> print(True/True)
1.0
>>> print(True//True)
1
>>> print(False//True)
0
>>> print(True//False)
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
ZeroDivisionError: integer division or modulo by zero
>>>
```